CS 556 Test 1

This is a closed-book exam, meaning no textbooks or electronic devices are allowed, except for one sheet of paper with handwritten notes. Write all your work, not only the final answer.

Question 1 *(20 points)* Your answer to any of the following questions should be 'insufficient information' if you feel the question does not supply enough information to determine the correct answer. Your answer can also be 'does not exist' if you judge that the question is about a property that is not applicable to the matrix in question. Given that a square matrix of dimension 45 has a linearly independent set of vectors (i.e., it is full rank), answer the following questions.

1. What is the rank of the matrix?
2. What is the numerical value of the determinant of the matrix?
3. What is the rank of the column space?
4. What is the dimension of the null space (i.e., nullity)?
5. What is the dimension of the left null space (i.e., nullity of the left null space)?
6. Is the matrix invertible (why or why not)?

Question 2 *(15 points)* Calculate the solution (if it exists) to the following system of linear equations (SLE) and demonstrate your methodology.

$$x_1 - \frac{4}{7}x_2 + \frac{2}{7}x_3 = \frac{11}{7}$$

$$-\frac{2}{7}x_2 + \frac{1}{7}x_3 = \frac{2}{7}$$

$$\frac{9}{7}x_2 + \frac{6}{7}x_3 = \frac{33}{7}$$

Question 3 *(15 points)* Answer whether the following statements are True, False or if there is insufficient information to make the determination. If a, b, c are a set of linearly independent vectors, each in R^3 and none of which are the zero vector, which of the following statements are true about them?

1. a, b, c span the space R^3.
2. There exists a set of scalars $\lambda_1, \lambda_2, \lambda_3$ where $\lambda_1 \neq \lambda_2$ which can be combined to form $\lambda_1 a + \lambda_2 b + \lambda_3 c = 0$.
3. The determinant of the matrix with a, b, c as columns is 0.
4. Consider a specific instantiation of the vectors as a $= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, b $= \begin{bmatrix} -1 \\ 4 \\ 9 \end{bmatrix}$, c $= \begin{bmatrix} -11 \\ 10 \\ 21 \end{bmatrix}$. Could this represent the set of vectors described in the original question?

Question 4 *(10 points)* Calculate the inverse of 2×2 matrix $A = \begin{bmatrix} -6 & 2 \\ 9 & -3 \end{bmatrix}$.

Question 5 *(15 points)* Employ Gram-schmidt orthogonalization to derive an orthonormal set of vectors from the set of independent vectors: a $= \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$, b $= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, c $= \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$.

Question 6 *(10 points)* Consider a vector $a_e = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ defined on the natural basis. Now, given the new basis defined by a matrix $B = \begin{bmatrix} 2 & 2 \\ -2 & 1 \end{bmatrix}$ how is the vector a_e expressed in the new basis?

Question 7 *(10 points)* Find all the eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$.

Question 8 *(5 points)* Find the best fit line using least squares approximation for the points (1,2), (2,1), (3,1) in R^2.